CNENGR202 Instrumentation and Control

Part 3 L1 – Introduction

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Embedded Systems

- At the end of this aspect of the module you should:
  - Understand the concept of the stored program microprocessor;
  - Understand the architecture and limitations of the basic microcontroller and how these can be overcome;
  - Be able to write, test and debug simple programs for a controller in Arduino;
  - Understand the concepts of subroutines and interrupts and how to use them;
  - Be able to use the principal on - chip facilities such as the timers and ADC.

Assessment -1

- You should prepare a report covering what you have achieved in the four laboratory sessions taken in weeks 14 and 16. Think of this in the context of condensing your knowledge, achievements and results into the report constraint. In particular, the report should be:

  - Technically accurate and consistent with the documentation that accompanies the Atmel 328P microcontrollers used in the labs
  - Attractive to the reader and accessible to all with an interest in the use of these devices

Assessment -2

- You should consider including
  - A very short piece on how and where microcontrollers are used
  - Brief details of the Atmel sufficient to set the context of the labs including its capabilities
  - An example programme showing and highlighting the important elements and a description of the development cycle using Atmel Studio
  - Consider including an example simple circuit diagram

- Of a professional Engineering standard.
- A complete record of what you have achieved in the labs

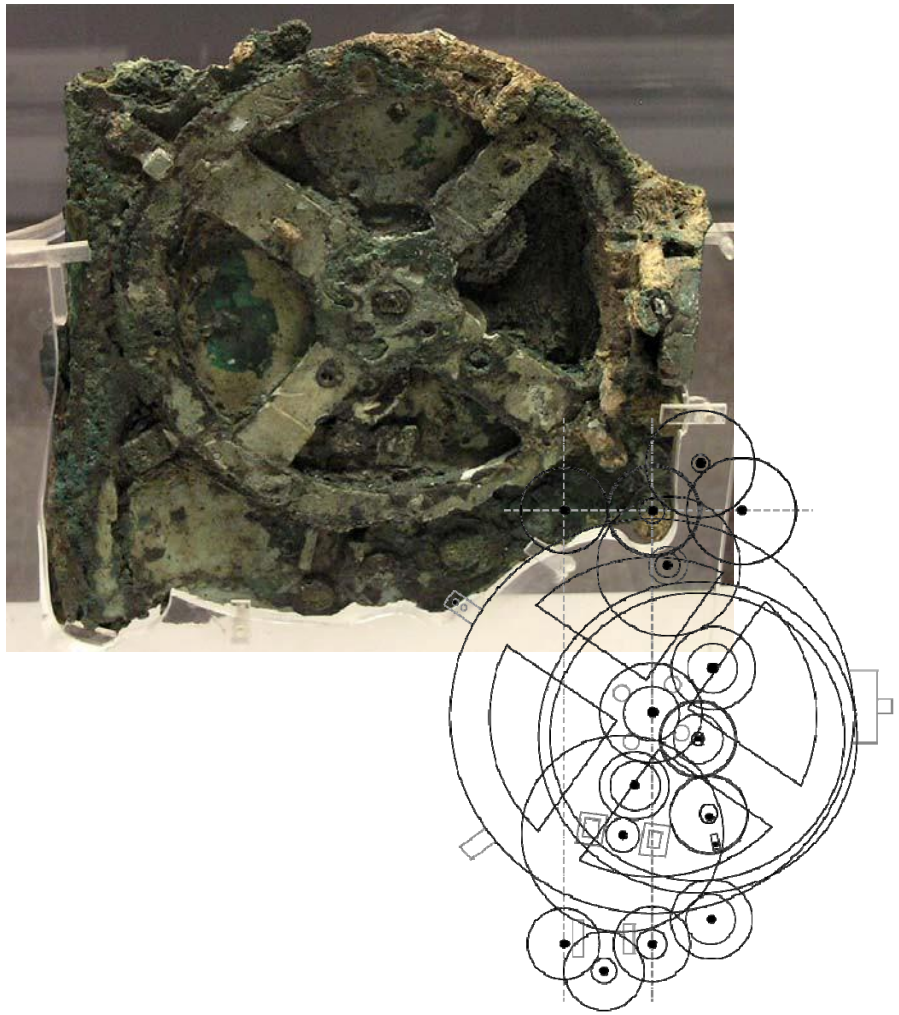
Recommended texts

- The ATMega328p Datasheet
  - https://ww1.microchip.com/downloads/en/DeviceDoc/ATmega48A-PA-88A-PA-168A-PA-328-P-DS-DS40002061B.pdf (also download from moodle)
  - https://ww1.microchip.com/downloads/en/DeviceDoc/Atmel-7810-Automotive-Microcontrollers-ATmega328P_Datasheet.pdf (also download from moodle)

- The ATMega328p Instruction Set Manual
  - https://ww1.microchip.com/downloads/en/DeviceDoc/AVR-Instruction-Set-Manual-DS40002198A.pdf (also download from moodle)

Programmable systems - Some history

- Self - operating machines are mentioned in many old texts from Greece, China, etc.
- An example is the **Antikythera mechanism**, an ancient analogue computer designed to calculate lunar and planetary motion, dated to the early 1st century BCE.
- Recovered c1901 from a wreck, its significance was not discovered for another century.
- Technology of this complexity did not appear again in Western Europe till the 14th century CE.



Programmable systems – Some history

- **Automata** - self operating machine, non - electronically controlled, especially those made to resemble human or animal actions, e.g. cuckoo clock.

- Al - Jazari (1136–1206) from modern turkey who constructed a hand - washing automaton that is employed the flush mechanism now used in modern toilets.



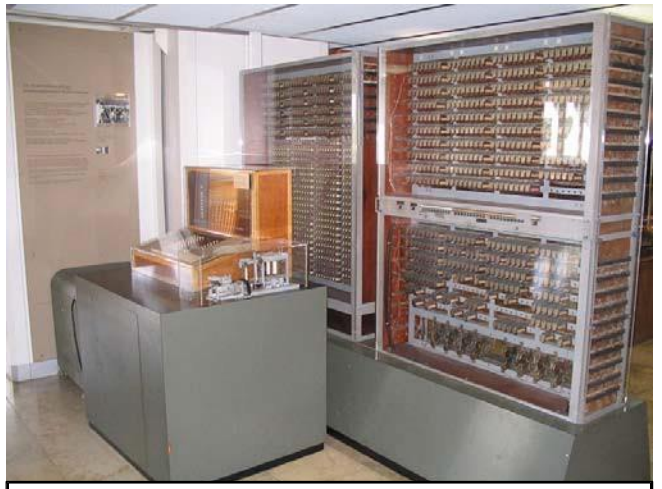
Programmable systems – Some history

- **Punched Cards** for controlling a sequence of operations undertaken by the machine.
- Machine used in controlling textile looms in France in 17th century



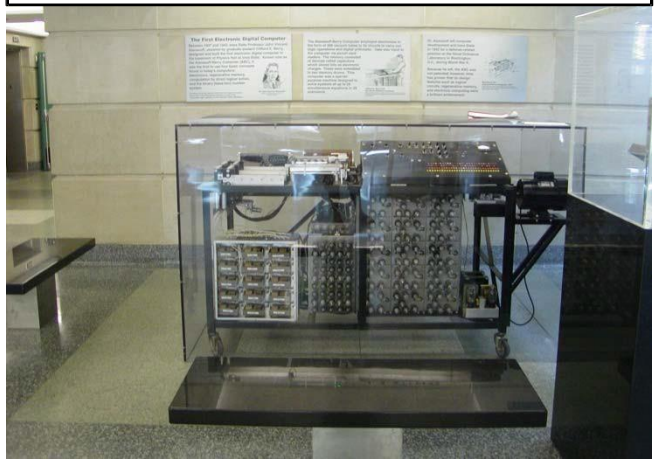
Programmable systems – Some history

- *Konrad Zuse* designed the electromechanical Z3 in 1941.
- It was the first working automatic digital computer featuring binary arithmetic (with floating point arithmetic) and some programmability.
- It had 2000 relays, and used a 22 - bit word length. Program code and data were stored on punched film.
- The non - programmable Atanasoff–Berry Computer (1941) used thermionic valves for computation with binary numbers.
- It had a regenerative capacitor memory, making it more compact than contemporary machines.



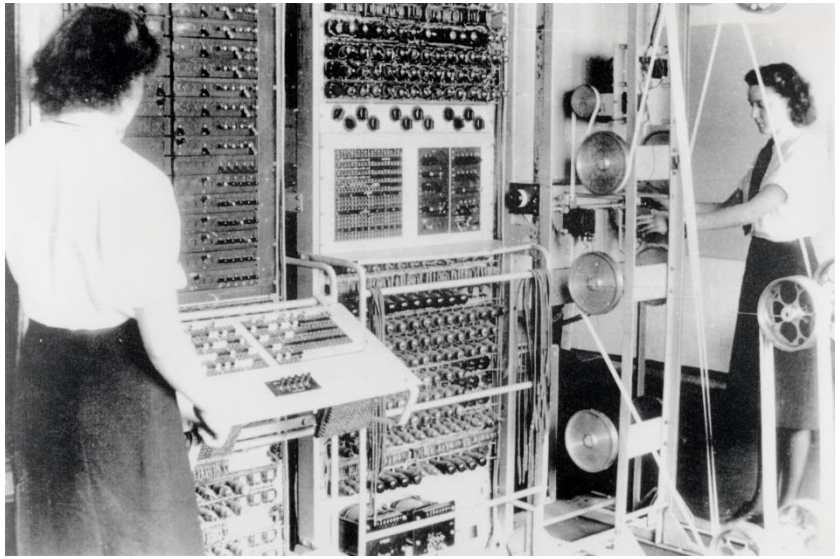
Zuse Z3 replica in Deutschen Museum in München - Venusianer

Atanasoff-Berry Computer replica at Durhum Center, Iowa State University - User:Manop



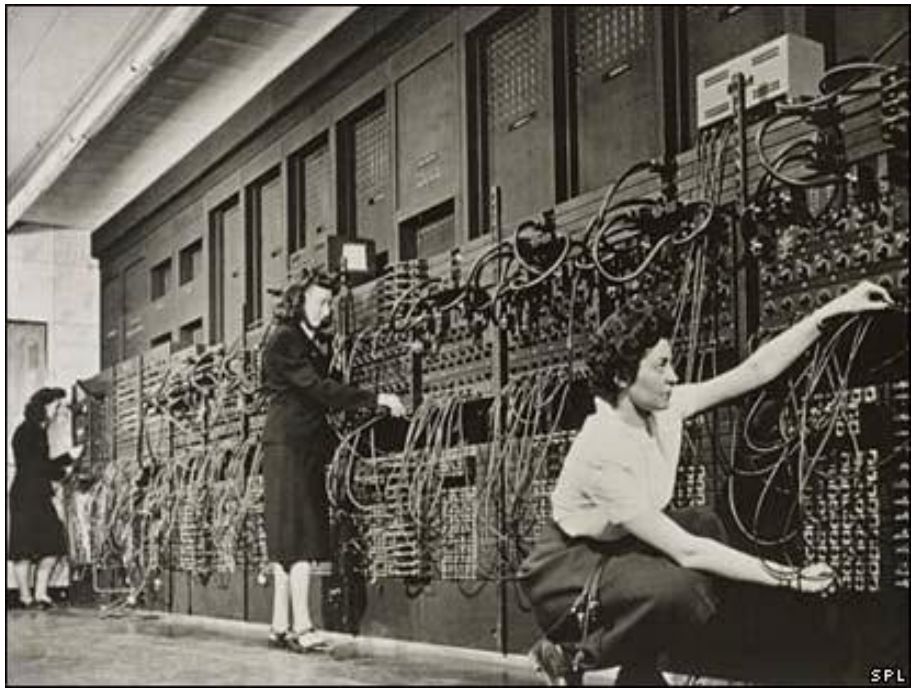
Programmable systems – Some history

- The British Colossus computers developed by codebreakers (1943)
- Used thermionic valves to perform Boolean and counting operations
- Generally regarded as worlds first programmable electronic and digital computer
- Programmed by switches and plugs as opposed to stored programs
- Used to help decipher and encrypt messages



Programmable systems – Some history

- J. Presper Eckert and John Mauchly of the Moore school of the University of Pennsylvania and the U.S. Army's Ballistic Research Laboratory built the Electronic Numerical Integrator and Computer ENIAC (1946)
- The first general purpose electronic computer.
- It used decimal arithmetic
- Initially required rewiring a plug board to change the programming



Programmable systems

- Von Neumann machine
- Fast forward to now, emergence of smart devices

…

Application to Embedded Systems -I







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Application to Embedded Systems - II







Introduction to Embedded Systems - I

What is an embedded system?

…any device that includes a programmable computer but is not itself a general - purpose computer (e.g. laptops, desktop).

Introduction to Embedded Systems - II

- Why do we need it?
- If we wish to control an Engineering system with a computer we have two choices:

External Computer Control

Engineering System



Embedded Computer Control

Engineering System

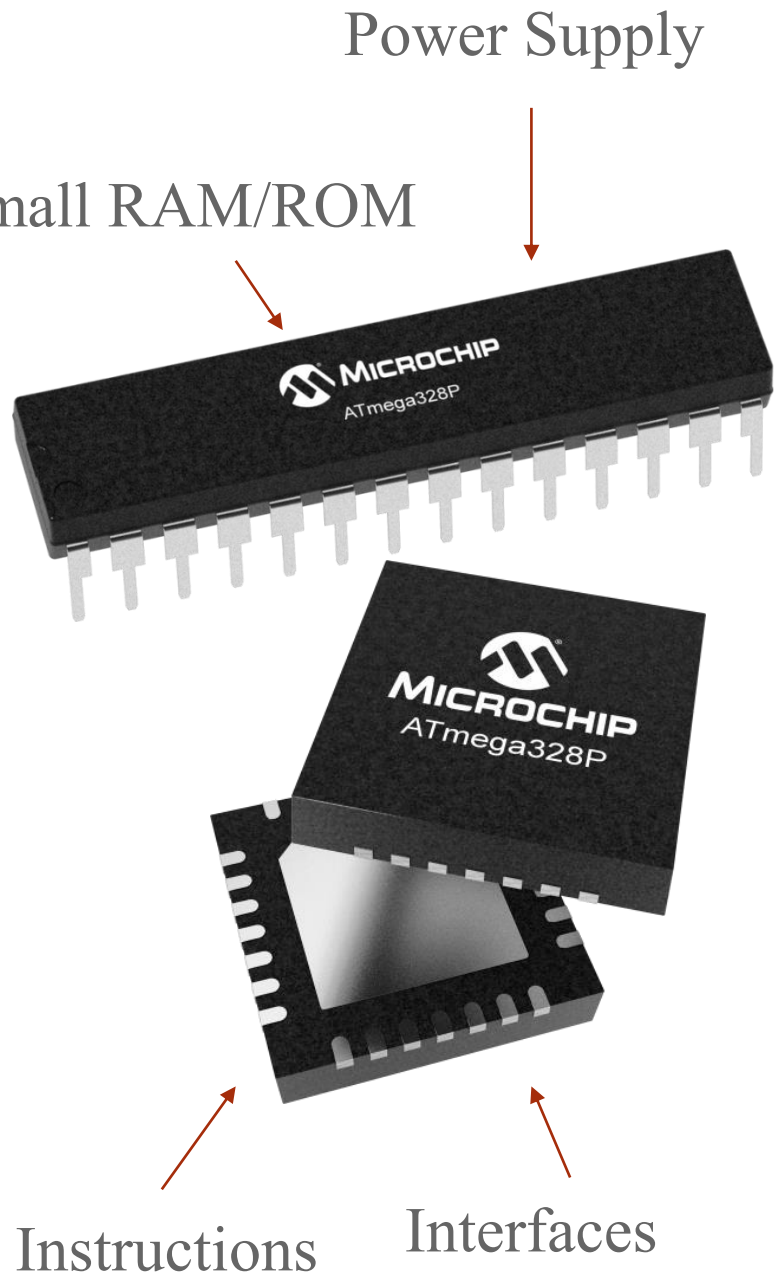


Microcontrollers

- Myriad(无数) of device technologies eg DSP, FPGA, microprocessors etc
- We will concentrate on Microcontrollers since these are most common for the control of Mechatronic systems. They are used in:
  - Automotive - engine management, climate control, instrumentation.
  - Industrial control - pumps, valves, solenoids, motors.
  - Telecommunications - display control, timing, encryption.
  - Medical - analysers, incubators, organ machines.
  - Consumer electronics - videos, microwaves, cookers, stereos, central heating controls.

Microcontrollers

- Microcontrollers are tiny computers in which:
  - There is a low power microprocessor
  - All ancillary features are often not integrated
    - power supply, I/O interfaces, clock etc.
  - Features for timing, analogue interface, multiplexing
  - There is very little memory for
    - Programs
    - Calculations
  - They are programmed externally



PC or Microcontroller?

External PC Control

- Pros
  - We have infinite flexibility about the scope of control.
- Cons
  - It is expensive
  - Fragile
  - Cumbersome
  - There are many redundant facilities and capacity
  - It is not suited to mass production
  - Security Issues

Microcontroller

- Pros
  - Cost - effective
  - Small, low - power
  - Reliable
  - Mass - produced
  - Easily serviceable
- Cons
  - There are constraints in terms of memory capacity and speed
  - Programming can be more complex

Arduino

- Arduino is an open‐source electronics prototyping platform based on flexible, easy‐to‐use hardware and software.
- It's intended for artists, designers, hobbyists, and anyone interested in creating interactive objects or environments.

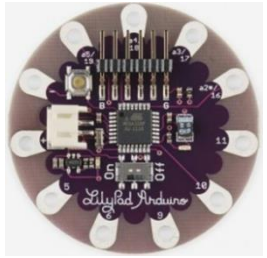
Why Arduino?

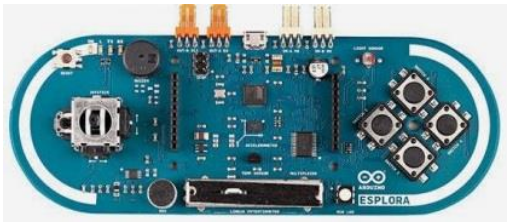
- Arduino microcontrollers have become the de facto standard.
- Many projects using Arduino microcontrollers.
- Strives for the balance between ease of use and usefulness.
- Inexpensive.

Arduino Boards

- Many different versions
  - Number of input/output channels
  - Form factor
  - Processor

- Leonardo
- Due
- Micro
- LilyPad
- Esplora
- Uno











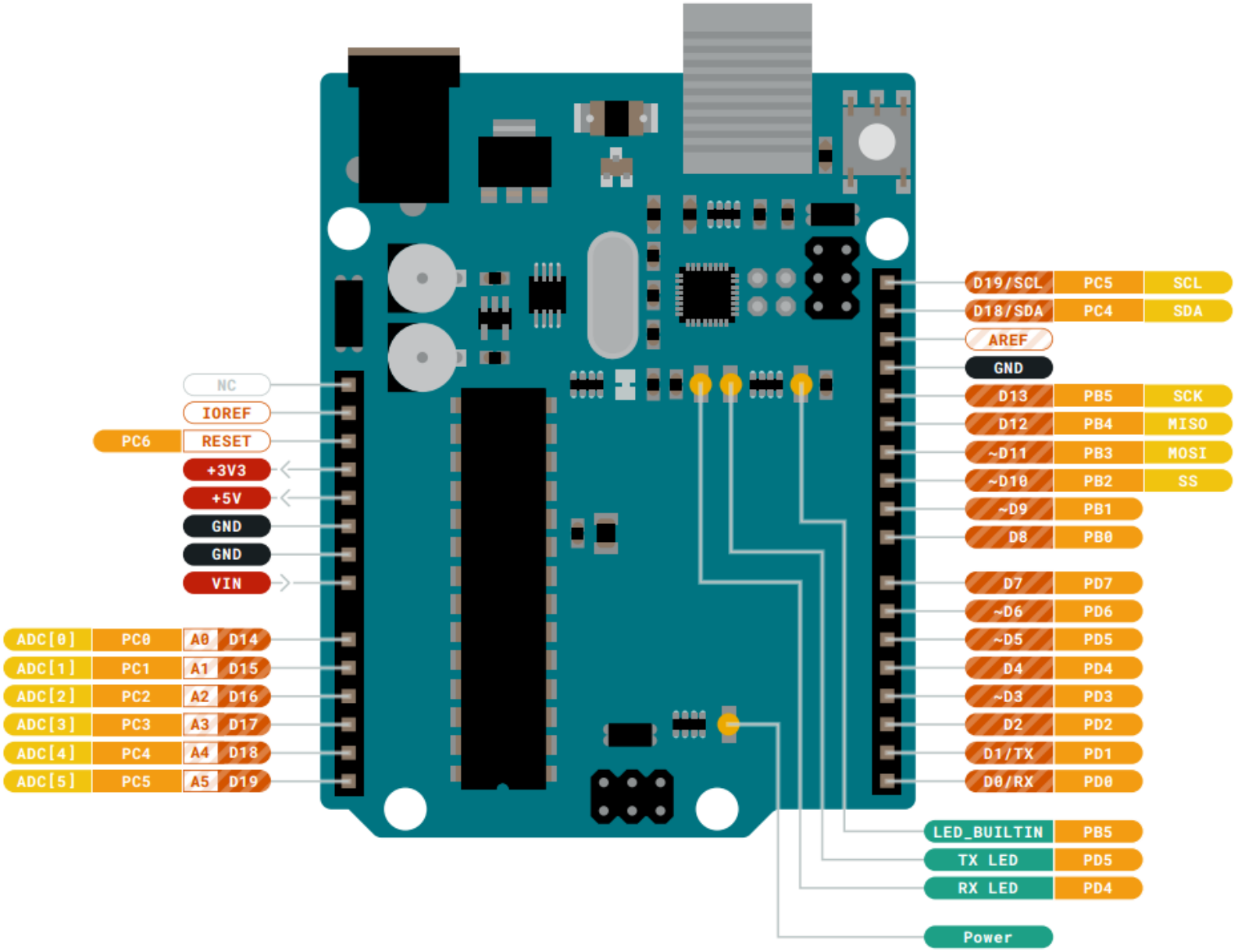


Arduino Uno - I

- Anatomy

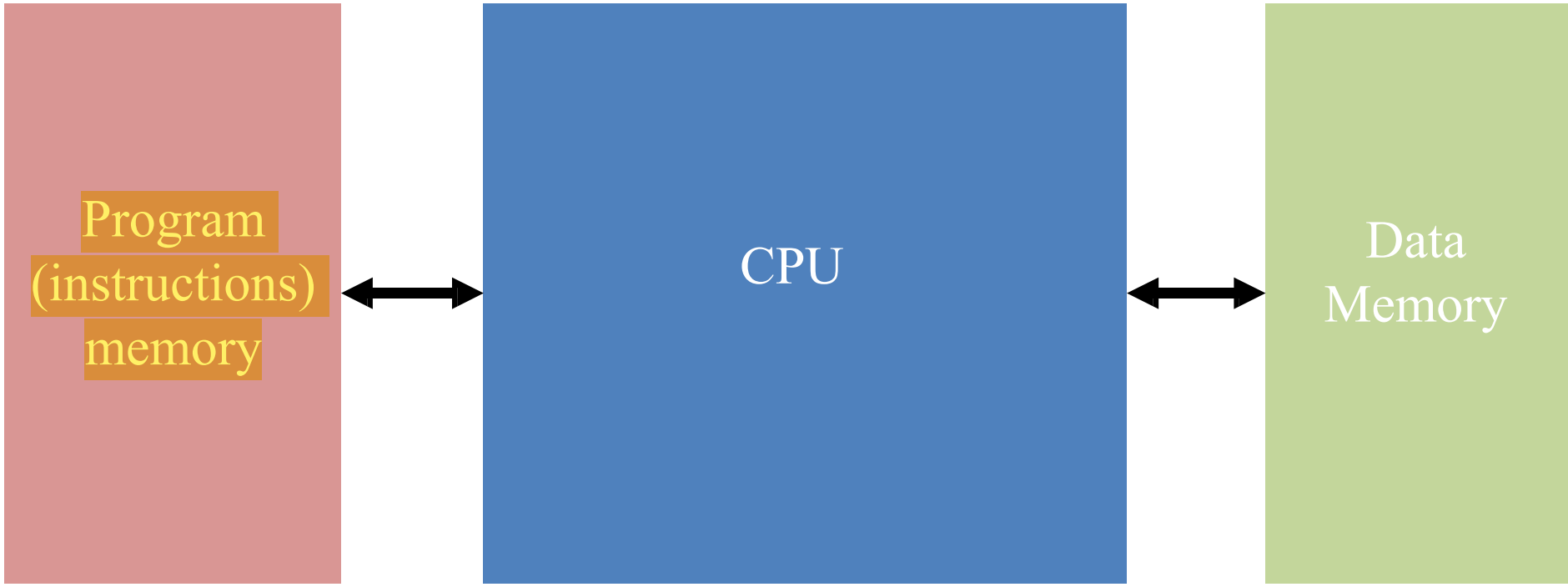


Arduino Uno - II



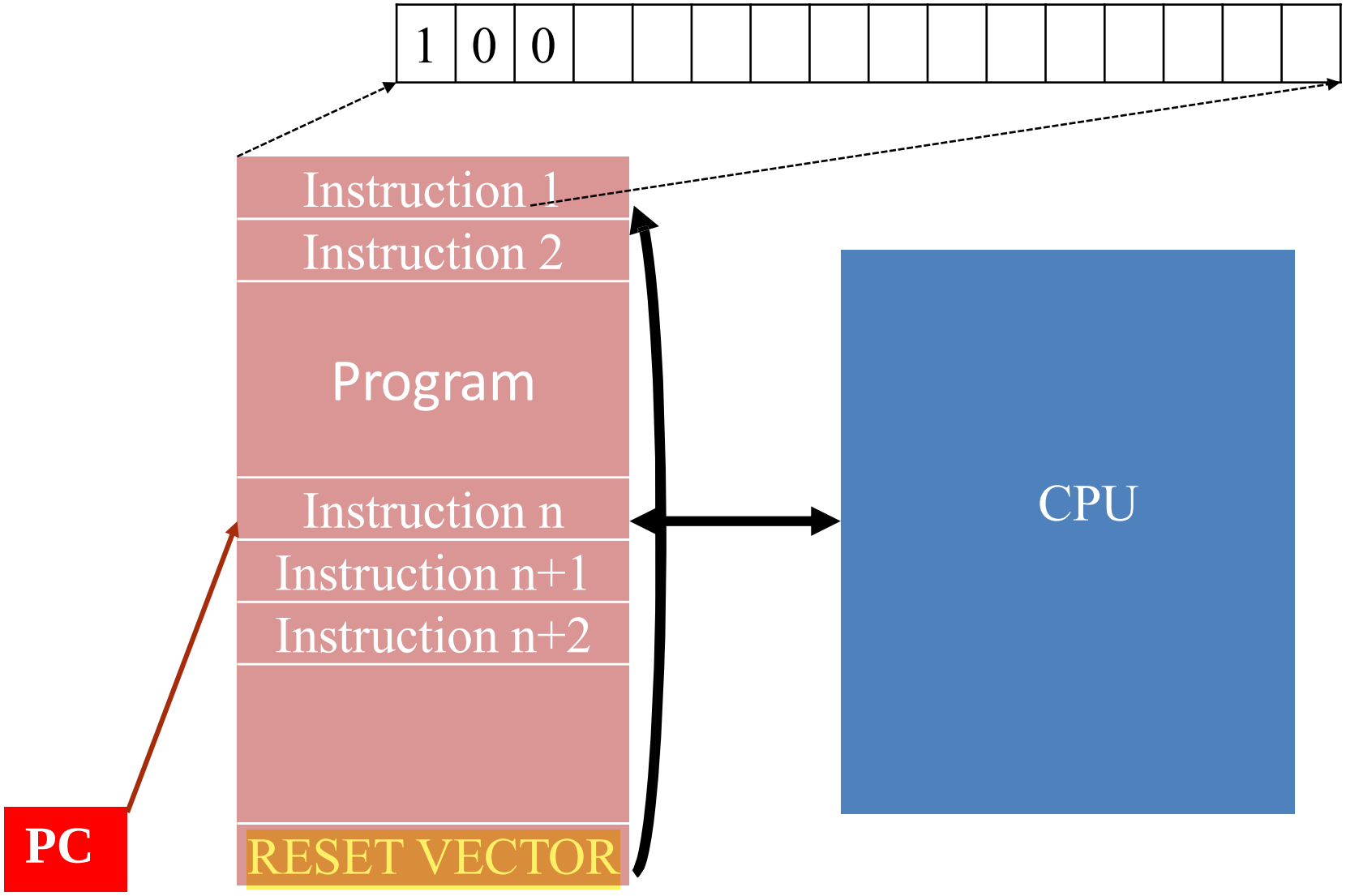
Programing the chip

Harvard Architecture



Programming

- The program memory stores the instructions
- Instructions are 16 - bit numbers
- Tell the CPU what to do
- Program Counter (PC) contains position of the current instruction



Instructions

- Represented here as hexadecimal numbers
- Bit in bold is the only bit that is an instruction
- Nasty looking, hard to understand

:	0	2	0	0	0	0	0	2	0	0	0	0	F	C
:	0	2	0	0	0	0	0	0	0	**E**	**E**	1	0	F
:	0	0	0	0	0	0	0	1	F	F				

Header

Program

End of file



Start code

Byte Count

Address

Record Type

Data

Checksum

Compiling

:	0	2	0	0	0	0	0	2	0	0	0	0	F	C
:	0	2	0	0	0	0	0	0	0	E	E	1	0	F
:	0	0	0	0	0	0	0	1	F	F				

Header

Program

End of file



Start code

Byte Count

Address

Record Type

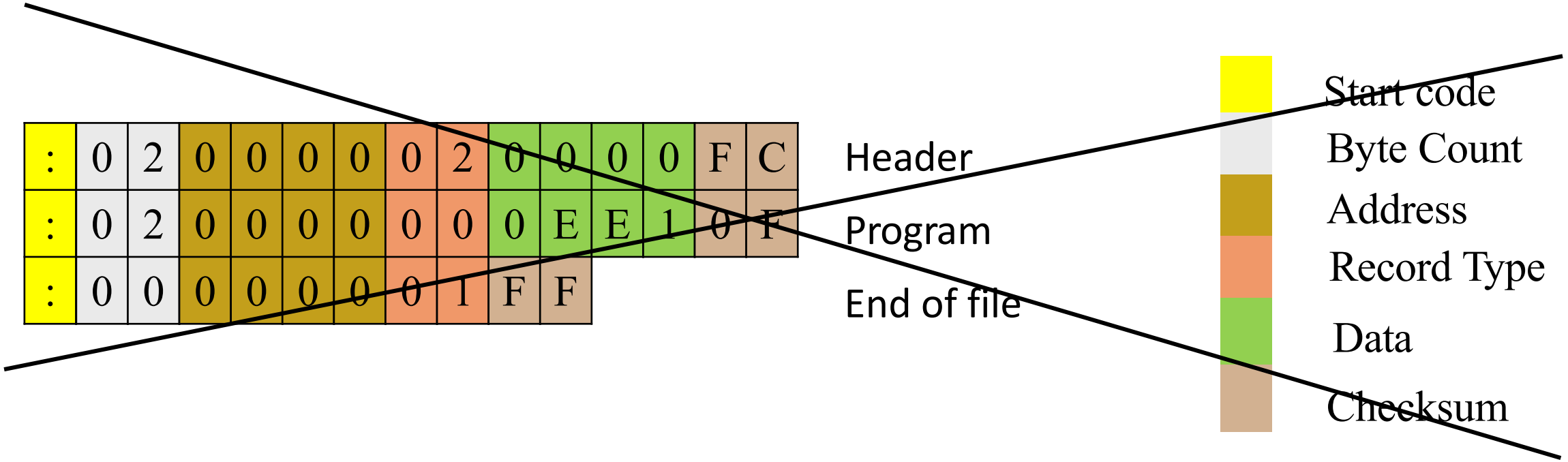
Data

Checksum

Compiles from:

LDI r16, 30 **OR** *int x = 30;*

Compiling



Compiles from:

LDI r16, 30 **OR** *int x = 30;*

What is compiler?

- Compiling is converting between a language and the instructions

LDI r16, 30 → Compiler →

:	0	2	0	0	0	0	0	2	0	0	0	0	F	C
:	0	2	0	0	0	0	0	0	0	E	E	1	0	F
:	0	0	0	0	0	0	0	1	F	F				

- Lower level than C, BASIC, Java etc.
  - Closer to the "metal"
  - Each C command may be many CPU instructions
- You will program in C, but lectures will cover both assembly and C

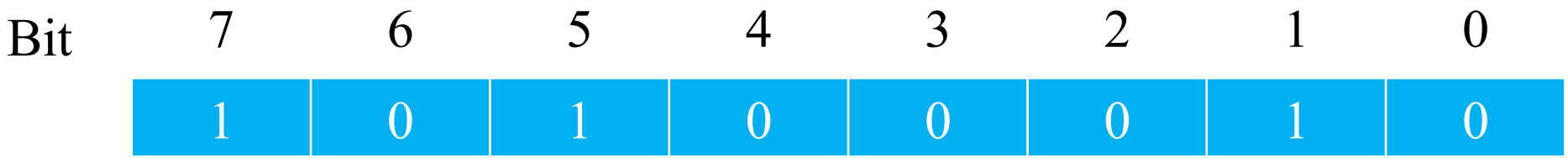
Inputting numbers

- A hex number is represented as 0xA2
- A binary number is represented as 0b10100010
- So you can have
  - *LDI r16*, 0xA2
- And
  - *LDI r16*, 0b10100010
- As well as
  - *LDI r16, 162*

- **It depends what is convenient**

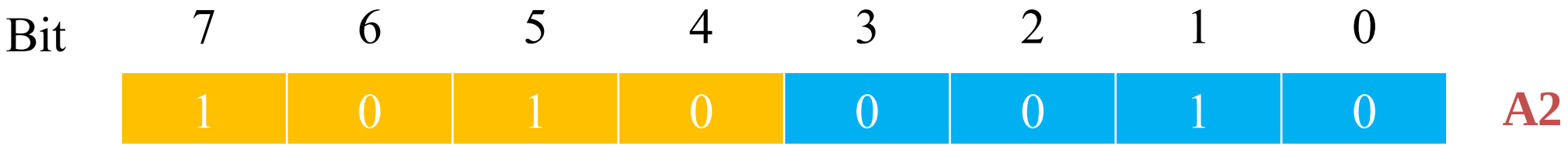
Revision of Hexadecimal

- A compromise is base - 16 or hexadecimal:



Revision of Hexadecimal

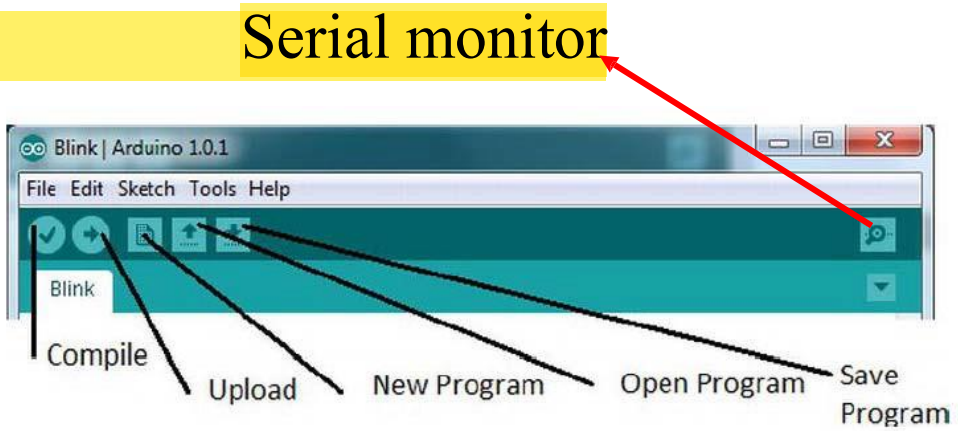
- A compromise is base - 16 or hexadecimal:
  - Split a byte into two 4 - bit nibbles and write the number as if these nibbles were separate;
  - Express as numbers of 16s and numbers of units;
  - Beyond '9' the letters A - F are used for 10 - 15;
  - The example below 'A2' is '162' in decimal.



Developing your program

IDE

- Arduino Integrated Development Environment (IDE)
- Simple, easy to use but no debuggers though
- Programs in Arduino are known as sketches
- Other IDE include AtmelStudio



Debugging

- No Debugger
- Use Serial.println() to on serial monitor observe output
- Remember to start in

```
setup(){

    Serial.begin(9600); // 9600 baud rate

}
```

波特率9600是什么意思　　波特率是一个电子信号上的术语，用于描述信道的数据传输速度。所谓信道，可以是无线的，也可以是有线的，说白了就是两个东西之间传输数据。

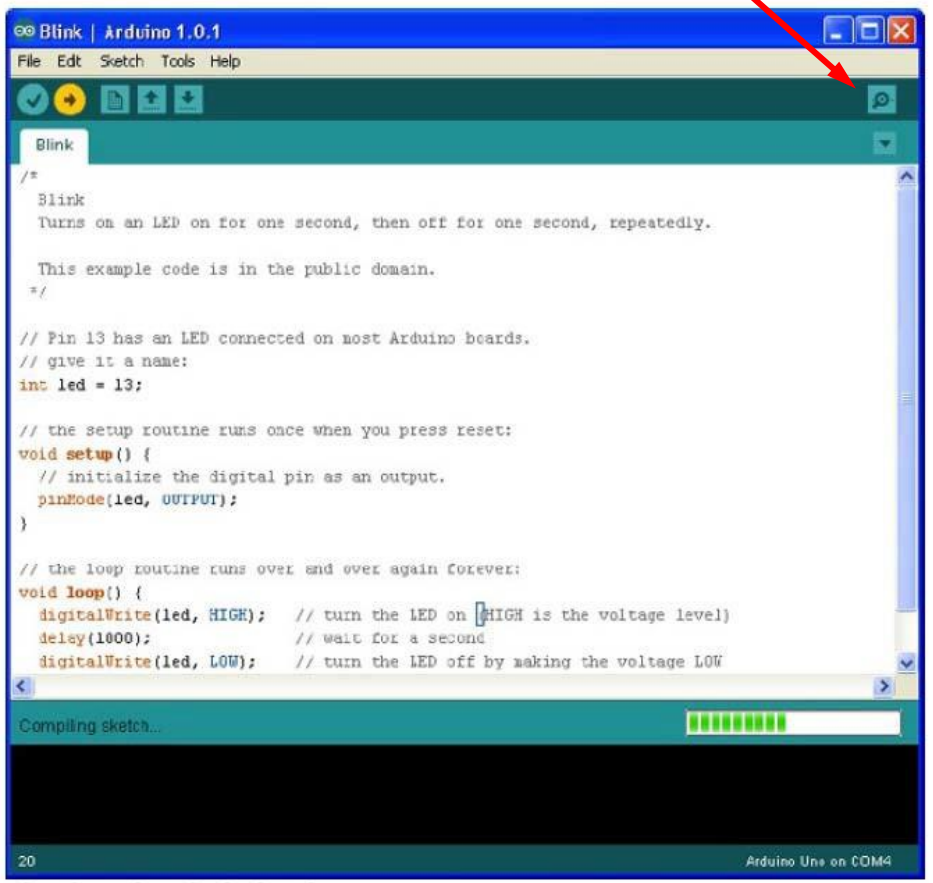
波特率通常单位是bit/s，**也就是** 二进制位/秒。因为一个字节是8**个**bit，而一个字节通常可以表达**一个**ASCII码，**比如一个英文字母，所以：**9600**的波特率的信道，**理论上每秒可以传输9200/8**个英文字母的数据量，也就是**1200个字节，大约1.2KB。**而**19200则是每秒可传输2400字节。

比如一个文件，假设容量是100KB，那么它通过9600的信道传输，需要的时间至少是100/1.2**秒，**也就是超过一分钟了。

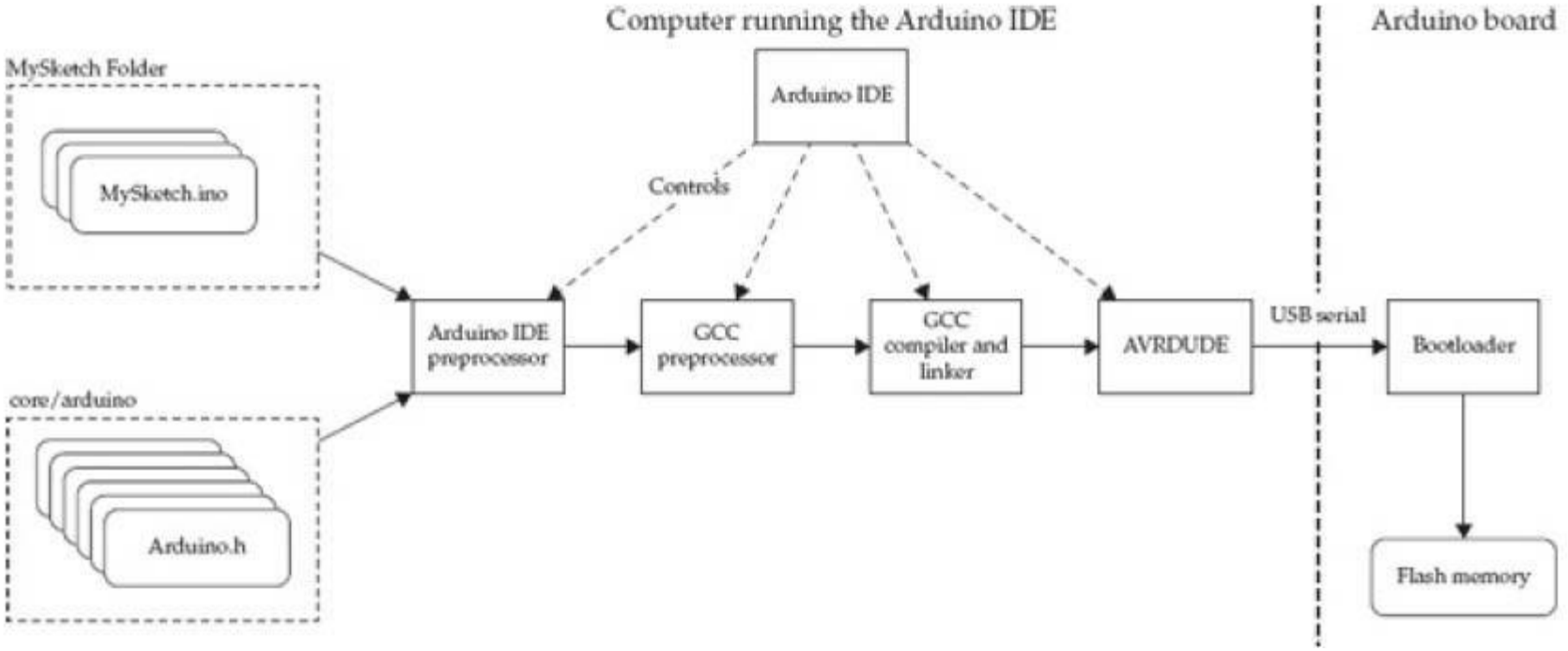
和19200**波特率的信道，**都是早期接口的传输率标准。比如早期的COM**口（也叫串口），**经常采用这样的标准，**今天看来，**速度非常非常慢。早期使用这样的接口用于鼠标、键盘等简单装置的连接，以及计算机和其它设备的通信。而现在，还有少数主板支持这样的接口。

在今天的计算机上，这个接口标准早已淘汰，但是生产中，还是有很多设备支持这样的接口。

Serial monitor

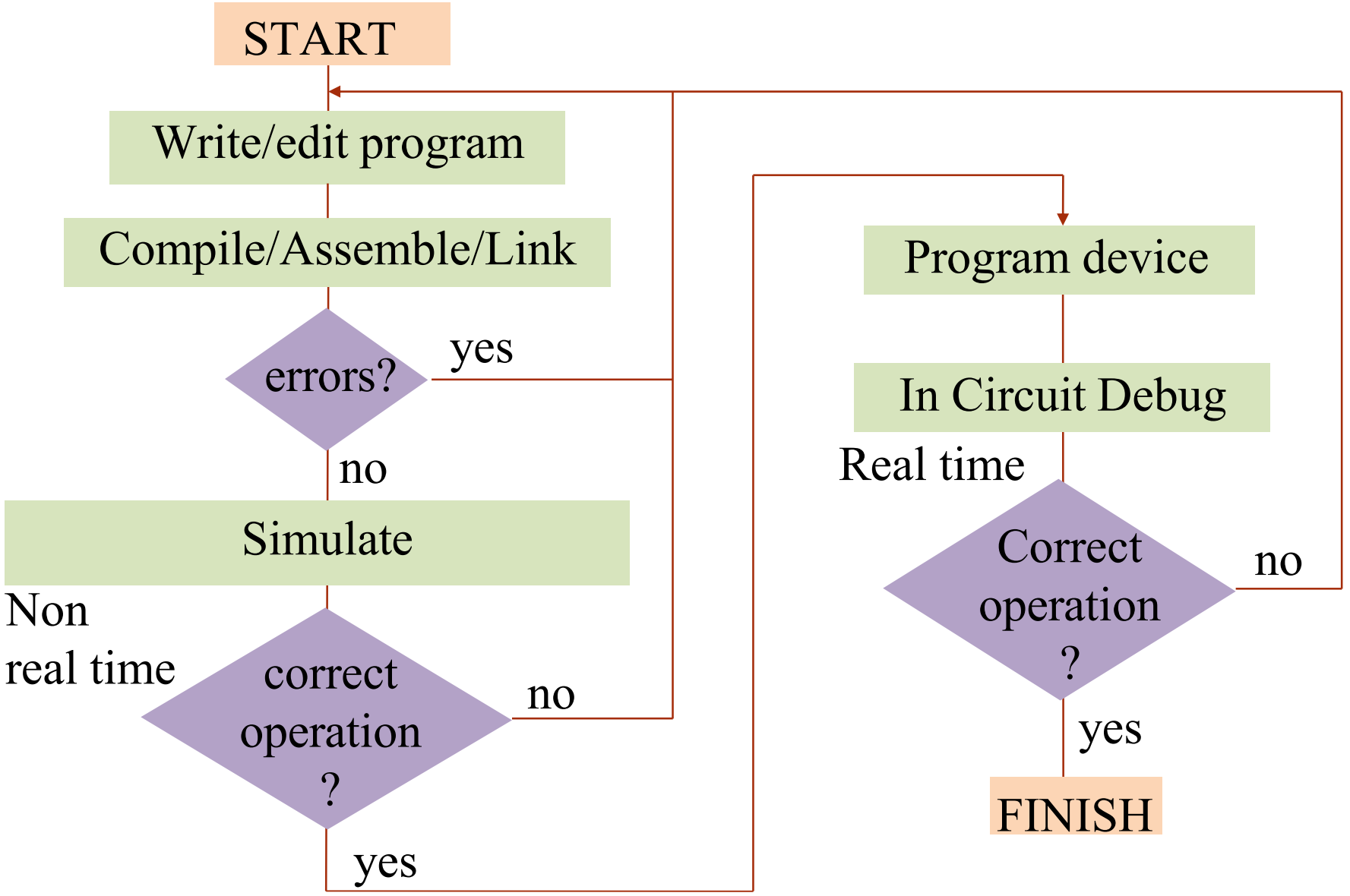


After hitting upload …



- Arduino sketches are in a text file with .ino extension
- Preprocessor assembles files that are part of the sketch
- Invoke open source C++ GCC Compiler to generate hexadecimal representation of machine code

Development Flow



Comments

- To annotate your code you should comment as much as possible
- To comment use ; or //

THIS is,code ;but this is a comment in assembly

SOME more,code //this is a comment in C

Summary

- Embedded systems all around us!
- Microcontrollers are used to control engineering systems
- Would like to control motors, leds, etc…
- Would like to read sensors and switches etc.